A Tableau-Based Water Quality Analysis with Machine Learning Approaches

1 Acknowledgements

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Abstract

This study evaluates water quality across various water bodies in Kerala using Tableau-based visual analysis and machine learning techniques. The dataset obtained from Kerala State Pollution Control Board was analysed to identify spatial patterns, district-wise variations, and the key factors influencing water quality. Water quality was assessed using the Indian Water Quality Index (IWQI) and the CCME Water Quality Index (CCME WQI). Physicochemical parameters, major ions, nutrients, and microbiological indicators were studied using correlation analysis and PCA for linear relationships, along with Mutual Information and Random Forest models to capture non-linear effects. In addition, monthly machine learning–based forecasts were developed for the Periyar and Bharathapuzha rivers for 2026, validated using 2025 data, and compared with Tableau's built-in forecasts, highlighting the usefulness of data-driven approaches for water quality management.

Keywords: Water Quality Index, Tableau, Machine Learning, Data Visualization, Multivariate Analysis

2 Introduction

Rivers and surface waterbodies are important for drinking-water supply, ecosystem health especially in regions like Kerala where surface water is a primary source of freshwater.Urbanization, land-use change etc have increased concerns regarding the degradation of water quality, making systematic assessment and monitoring essential for effective water resource management.

Water Quality Indices (WQIs) are widely used as integrated tools to summarize physicochemical and microbiological data into a single interpretable value. Among these, the Indian Weighted Arithmetic Water Quality Index is commonly applied for assessing drinking-water suitability.However, drinking-water-focused indices can be sensitive to parameters such as salinity, which may reflect natural conditions of water rather than direct pollution, particularly in coastal environments.To complement drinking-water assessments, the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI) has been widely adopted for evaluating general water quality.The CCME WQI provides a flexible framework that can be adapted to different objectives and parameter sets and offers insight into overall river health independent of salinity-related effects.For selected rivers, site-specific analysis was carried out to obtain a clearer and more detailed interpretation of water quality variations.

Water quality is shaped by a combination of natural geochemical processes and human activities such as sewage discharge, agricultural runoff, and changes in land use. These influences often act together, making it difficult to clearly interpret patterns in water quality data.To address this complexity, multivariate statistical techniques like correlation analysis and Principal Component Analysis (PCA) are widely used to explore relationships among water quality parameters and to simplify complex datasets. While these methods are effective in identifying general trends, they mainly describe linear relationships and may overlook important non-linear interactions that commonly occur in environmental systems. Machine learning approaches, including Mutual Information analysis and Random Forest models, help overcome this limitation by capturing non-linear dependencies and highlighting the relative importance of different parameters.In this study, statistical and machine learning methods are combined to analyse water quality data in a more comprehensive way, aiming to identify the key factors controlling water quality variability and to gain better insight into the hydrochemical and biological processes influencing the system.

We also analysed the physical properties of water like colour, temperature, total dissolved solids, total suspended solids and chemical properties like pH, dissolved oxygen, temporary hardness, nitrate content. This is critical to ensure safety for human consumption, maintaining aquatic ecosystems, optimizing industrial and agricultural use, water quality management.

The study further includes machine learning–based forecasting of water quality by predicting the WQI for two of the most important rivers of Kerala, namely the Periyar and Bharathapuzha. WQI time series are often nonlinear, irregularly sampled, noisy, and influenced by seasonal effects, making reliable forecasting challenging.

Conventional approaches such as linear regression or exponential fitting fails at such situation, However, recent developments in machine learning such as powerful tools for modeling nonlinear dependencies, handling missing data, and incorporating temporal memory through lagged features, has enabled more accurate water quality predictions.

3 Dataset Description & Collection Details

The dataset was obtained from the official Kerala Government website. We used the August 2025 records, which are available as part of the month-wise datasets provided at the Kerala State Pollution Control Board portal: [`https://kspcb.kerala.gov.in/nwmp`.]. It contains 182 monitoring records ; each record corresponds to one monitoring observation and includes information about the monitoring station, such as the station code, latitude and longitude, district, and river basin.The dataset consists of 53 parameters in total. Among these, 21 are numerical parameters, which include commonly measured water quality variables such as physicochemical properties and microbial indicators. The remaining 30 parameters are categorical or descriptive, providing location details, administrative classification, and site-level observations.The monitored water bodies are classified into the following types: river, lake, pond, canal, drain, beach, and sea. To better understand the coverage of the dataset, a visual listing of rivers included within each district is shown in 1. Water quality data were obtained for the years 2020-2025 of the rivers Periyar and Bharathapuzha for ML Based WQI forecast. Each dataset consists of monthly observations of multiple water quality parameters, including:

- Physico-chemical parameters such as pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), nutrients, hardness, and electrical conductivity
- Microbiological indicators including total coliform and fecal coliform counts


Rivers in each district

Alappuzha	ACHANKOVIL.. PAMBA.. VARATTAAR..
Ernakulam	CHITHRAPPUZHA.. KADAMBAYAR.. KOTHAMANGALAM.. MUVATTAPUZHA.. PERIYAR..
Idukki	PERIYAR..
Kannur	ANJARAKANDY.. KAKKAD.. KAVVAYI.. KUPPAM.. PERUVAMBA.. RAMAPURAM.. THALASSERY.. VALAPATTANAM..
Kasaragod	CHANDRAGIRI.. KARYANKODE.. MADAKKARA.. MANJESHWAR.. MOGRAL.. NEELESWARAM.. PULLUR.. SHIRIYA.. UPPALA..
Kollam	ITHIKKARA.. KAKKAD.. KALLADA.. PALLICKAL..
Kottayam	MEENACHIL.. MUVATTAPUZHA..
Kozhikode	CHALIYAR.. KALLAYI.. KORAPUZHA.. KUTTIYADY.. MAHE.. RIVER IRUVAZHINJI..
Malappuram	BHARATHAPUZHA.. CHALIYAR.. KADALUNDI.. TIRUR..
Palakkad	BHARATHAPUZHA.. BHAVANI.. CHITHRAPPUZHA.. KALPATHI PUZHA.. KANNADIPPUZHA.. KORAYAR.. RIVER KUNTHIPUZHA.. –
Pathanamthitta	ACHANKOVIL.. MANIMALA.. PALLICKAL.. PAMBA..
Thiruvananthap..	AYIROOR.. KARAMANA.. KILLIYAR RIVER.. MAMOM.. NEYYAR.. VAMANAPURAM..
Thrissur	CHALAKUDY.. KARUVANNUR.. KECHERI.. PERIYAR.. PUZHAKKAL.. WADAKKANCHERY..
Wayanad	KABANI.. KARAPUZHA.. MANIYANKODE PUZHA.. RIVER KALINDI.. RIVER PAPANASINI..

Figure 1: District-wise listing of recorded rivers

4 Methodology

4.1 Data Preprocessing

Values reported as below the detection limit (BDL) were replaced by half of the corresponding limit of detection (LOD/2), a commonly adopted approach in environmental studies to reduce bias associated with censored data.

4.1.1 Correlation Analysis

Physicochemical parameters, major ions, nutrients and microbial indicators were used for the analysis, but the non-chemical variables such as station codes, geographical coordinates and administrative details were excluded. The values of parameters that fall below the detection limit (BDL) were replaced by half of the corresponding detection limit (LOD/2). Missing values were handled using the K-neighbour (KNN) imputation method with $k = 5$, allowing the preservation of multivariate relationships within the data set. To make sure each parameter contributes equally we standardized selected variables using z-score normalization.

4.1.2 ML based WQI Forecast

To ensure temporal and multivariate consistency, missing numerical values were imputed using the **K-Nearest Neighbors (KNN)** imputation method.

KNN imputation was preferred over mean or median substitution as it preserves multivariate structure by estimating missing values based on similarity to neighboring observations.

Sampling dates were reported in multiple formats, including *Mon-Year*, *Year-Mon*, and *DD-MM-YYYY*. All date entries were converted to the same format for uniformity.

4.2 Water Quality Index Calculation

Water quality analysis was carried out using two indices: the Indian Weighted Arithmetic Water Quality Index (Indian WQI), focusing on drinking water suitability, and the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI), providing a general assessment of water quality.

4.2.1 Indian Weighted Arithmetic Water Quality Index

The Indian Water Quality Index was evaluated using the weighted arithmetic index method as applied to Indian rivers by Tiwari and Mishra 1985, following the application described by Jyothi et al. 2021.It was computed based on drinking-water standards prescribed by the ICMR and BIS. Eleven physicochemical parameters were considered(1). Each parameter was assigned a relative weight according to its importance in drinking-water quality assessment(2)

S. No.	Parameter	Permissible limit
1	pH	6.5–8.5
2	Electrical Conductivity (μS cm^{-1})	300
3	Total Dissolved Solids (mg L^{-1})	500
4	Total Hardness (mg L^{-1})	300
5	Calcium (as $CaCO_3$, mg L^{-1})	75
6	Magnesium (as $CaCO_3$, mg L^{-1})	30
7	Total Alkalinity (mg L^{-1})	120
8	Chlorides (mg L^{-1})	250
9	Nitrates (mg L^{-1})	45
10	Dissolved Oxygen (mg L^{-1})	5
11	Biochemical Oxygen Demand (mg L^{-1})	5

Table 1: Parameters and permissible limits used for the Indian Water Quality Index calculation

S. No.	Parameter	Unit weight (w_i)
1	pH	0.2190
2	Total Hardness	0.0062
3	Chlorides	0.0074
4	Electrical Conductivity	0.3710
5	Calcium	0.0250
6	Total Dissolved Solids	0.0037
7	Total Alkalinity	0.0155
8	Magnesium	0.0610
9	Dissolved Oxygen	0.3723
10	Nitrates	0.0412
11	Biochemical Oxygen Demand	0.3723

Table 2: Unit weights of Indian water quality parameters

For each parameter, a quality rating (q_i) was calculated as

$$q_i = 100 \times \frac{V_i - V_{\text{ideal}}}{V_{\text{standard}} - V_{\text{ideal}}}, \tag{1}$$

where V_i is the observed value, V_{standard} is the permissible limit, and V_{ideal} is the ideal value in clean water.The overall WQI was obtained as the weighted sum of individual sub-indices.Water quality classification was carried out following the Indian Weighted Arithmetic WQI methodology as described in Jyothi et al. 2021:

- **Excellent quality water:** WQI 0–25
- **Good quality water:** WQI 26–50
- **Poor quality water:** WQI 51–75
- **Very Poor quality water:** WQI 76–100
- **Unfit for drinking:** WQI >100

4.2.2 Canadian Council of Ministers of the Environment Water Quality Index

The water quality index was calculated using Canadian Council of Ministers of the Environment 2001. Five parameters-pH, dissolved oxygen, biochemical oxygen demand, total coliform, and fecal coliform were used. Except for fecal coliform, drinking water standards prescribed by the Bureau of Indian Standards (BIS) and the Indian Council of Medical Research (ICMR) were applied 3. For fecal coliform, the standard was taken from the Central Pollution Control Board (CPCB) primary water quality criteria for surface waters, using the desirable limit of 500 MPN/100 mL to better capture microbiological contamination.

Parameter	**Standard Value**
pH	6.5–8.5
Dissolved Oxygen (mg/L)	≥ 5.0
Biochemical Oxygen Demand (mg/L)	≤ 3.0
Total Coliform (MPN/100 mL)	≤ 500
Fecal Coliform (MPN/100 mL)	≤ 500

Table 3: Parameters and Standard values used in CCME WQI

Water quality Index (WQI) was calculated using the scope (F_1), frequency (F_2), and amplitude (F_3) components following the methodology proposed by the Canadian Council of Ministers of the Environment.Canadian Council of Ministers of the Environment 2001. An explanation of each factor and how they are combined to determine water quality is provided below.

The scope factor (F_1) represents the percentage of water quality parameters that do not meet their respective objectives and is given by

$$F_1 = \frac{\text{Number of failed parameters}}{\text{Total number of parameters}} \times 100. \tag{2}$$

The frequency factor (F_2) denotes the percentage of individual tests that fail to meet water quality objectives and is defined as

$$F_2 = \frac{\text{Number of failed tests}}{\text{Total number of tests}} \times 100. \tag{3}$$

The amplitude factor (F_3) provide a measure for extent to which failed test values deviate from their objective. For each failed test, an excursion value is calculated. If the measured value exceeds the objective,

$$\text{excursion}_i = \frac{\text{Failed value}_i}{\text{Objective}_i} - 1, \tag{4}$$

and if the measured value is below the objective,

$$\text{excursion}_i = \frac{\text{Objective}_i}{\text{Failed value}_i} - 1. \tag{5}$$

The normalized sum of excursions (nse) is computed as

$$nse = \frac{\sum \text{excursion}_i}{\text{Total number of tests}}, \tag{6}$$

and the amplitude factor is obtained as

$$F_3 = \frac{nse}{0.01\, nse + 0.01}. \tag{7}$$

Although the CCME framework recommends a minimum of four parameters measured at least four times, the present analysis is based on single observations per parameter.Accordingly, both the scope (F_1) and frequency (F_2) components were calculated based on the proportion of parameters exceeding their respective standard values. Finally, the CCME Water Quality Index is calculated as

$$\text{CCME WQI} = 100 - \sqrt{\frac{F_1^2 + F_2^2 + F_3^2}{3}}. \tag{8}$$

According to Canadian Council of Ministers of the Environment 2001, water quality is classified as follows:

- **Excellent:** 95–100
- **Good:** 80–94
- **Fair:** 65–79
- **Marginal:** 45–64
- **Poor:** 0–44

4.3 Tableau Visualizations

4.3.1 Spatial and Waterbody-wise Assessment of Water Quality in Kerala

To understand how the dataset is distributed, a geospatial analysis of waterbodies across Kerala's districts was performed.District-level aggregation was first applied to count distinct waterbodies, and then monitoring locations were aggregated within each district.

Water quality index was evaluated using the Indian Weighted Arithmetic Water Quality Index. For waterbodies with multiple monitoring locations,WQI values were averaged to obtain a representative index for each waterbody. Based on these values, waterbodies were classified into standard water quality categories, and the distribution across categories was summarized in tabular form. The same procedure was applied for the CCME WQI, which provided a complementary assessment less sensitive to salinity-related parameters.

To analyze spatial variability along individual river stretches, four representative rivers were selected for detailed analysis based on Indian WQI results: Bharathapuzha and Karamana Rivers (classified as Poor), Thalassery River (Unfit for Drinking), and Pamba River (Good). For these rivers, WQI values from multiple monitoring locations were analyzed using both Indian and CCME indices.

4.4 Machine Learning

4.4.1 Correlation Analysis

We used Pearson's correlation analysis to look at the relationships among selected water quality parameters. The correlation coefficients were compiled into a correlation matrix and visualised using a heat map, to help identify both strong and weak associations. To further explore the underlying structure of the dataset, Principal Component Analysis (PCA) was applied to the data. PCA reduced the dimensionality of the dataset and helped identify groups of interrelated parameters. The components with eigenvalues greater than one were considered as significant. We used the scree plot to help us assess the explained variance and determine the appropriate number of components to retain. We then performed Hierarchical Cluster Analysis (HCA) on the data and PCA loadings to group chemically similar parameters and to support the associations identified through PCA.

4.4.2 Dependency-Based Machine Learning Analysis

In addition to linear correlations, dependency relationships among water quality were examined using Mutual Information (MI) analysis, which is capable of capturing non-linear associations. Furthermore, a Random Forest regression model was used to evaluate the importance of each individual parameter. We used the composite water chemistry index as the proxy target. The variable for this analysis.

4.5 Physical and Chemical Analysis of River water

Colour: Analysed the colour of rivers in all districts to find which colour river is more prevalent in each district. White indicates clear rivers, green indicates rivers with algae content, violet indicates turbid rivers with suspended solids (silt, clay, plankton, debris) and microorganisms, and brown indicates a high amount of suspended sediments and organic matter.

Temperature: Plotted the average temperature of rivers. Normal river water temperature is 0° to 25°C. High temperature in river water decreases the dissolved oxygen concentration, causing stress for aquatic life.

Total Dissolved Solids: TDS (Total Dissolved Solids) measures the concentration of dissolved inorganic salts (calcium, magnesium, potassium, sodium, bicarbonates, chlorides, sulfates) and small amounts of organic matter in river water. Lower levels (< 300 mg/L) represent fresher, less mineralised water. Very low levels (< 50 mg/L) can taste flat or be corrosive, while high levels (> 500 mg/L) can cause metallic taste, hardness, and potential health concerns. Plotted the average TDS amount in rivers across districts.

Total Suspended Solids: Total Suspended Solids (TSS) are tiny, non-dissolved particles (like silt, algae, plankton, organic matter) floating in water. Plotted the average TDS amount in rivers across districts.

pH: pH is a scale used to measure the acidity or basicity of a solution, with values ranging from 0 to 14. A pH of 7 is neutral, a pH below 7 is acidic, and a pH above 7 is basic or alkaline. Analysed the average pH level in the rivers of each district.

Dissolved Oxygen: Dissolved oxygen is a physico - chemical property. We plotted the average dissolved oxygen content in rivers across districts.

Temporary Hardness: Temporary hardness in water is caused by dissolved calcium and magnesium bicarbonates (like $Ca(HCO_3)_2$), making it "hard" because it reacts poorly with soap, but it's "temporary" because it can be easily removed by simple boiling. Plotted the average amount of calcium and magnesium in all rivers of each district. And also the temporary hardness of water caused by these dissolved calcium bicarbonate and magnesium bicarbonate.

Nitrate: Analysed the relationship between the colour of rivers and their nitrate content. Generally, healthy, unpolluted rivers have concentrations between 0.1 mg/L and 2.2 mg/L (as nitrogen).

Biochemical Oxygen Demand (BOD): Biochemical Oxygen Demand is a key indicator of organic pollution in surface waters, representing the amount of dissolved oxygen required by aerobic microorganisms to decompose biodegradable organic matter. Elevated BOD levels indicate increased organic loading from sources such as domestic sewage, agricultural runoff, and industrial effluents, and can lead to oxygen depletion and stress on aquatic ecosystems. In this study, average BOD concentrations were analyzed across rivers to assess spatial variability in organic pollution.

4.6 ML-based WQI Forecast

To capture system memory, persistence, and delayed responses in water quality dynamics, lag-based temporal features were constructed:

- **WQI_Lag1**: WQI value from the previous month

- **WQI_Lag2**: WQI value from two months prior
- **WQI_Gradient**: First-order temporal change in WQI

These features allow machine learning models to learn patterns of persistence, recovery, and degradation in river water quality.

Seasonal variability was encoded using cyclic transformations of the month variable to preserve periodic continuity:

$$\text{Month_Sin} = \sin\left(\frac{2\pi m}{12}\right), \quad \text{Month_Cos} = \cos\left(\frac{2\pi m}{12}\right) \tag{9}$$

where m denotes the month of the year. This representation avoids artificial discontinuities between December and January and enables smooth learning of seasonal patterns by machine learning models.

To model and forecast the Water Quality Index (WQI), five regression-based machine learning models were evaluated:

- **Extreme Gradient Boosting (XGBoost)** with a Tweedie objective function, chosen for its robustness to skewed and non-negative target distributions
- **Random Forest Regression**, which uses ensemble averaging to capture nonlinear relationships and reduce overfitting.
- **Gradient Boosting Regression**, based on sequential error correction to improve predictive accuracy
- **Support Vector Regression (SVR)**, utilizing kernel-based methods to model complex nonlinear dependencies
- **Linear Regression**, used as a baseline linear model for comparison

The dataset was split as:

- **Training data**: observations up to and including 2024
- **Testing data**: observations from 2025

To provide a meaningful baseline for comparison, a persistence model was used, defined as:

$$\text{WQI}_{t+1} = \text{WQI}_t \tag{10}$$

Model performance was evaluated using the **Root Mean Squared Error (RMSE)**. In addition, a skill score was computed to quantify improvement relative to the persistence baseline:

$$\text{Skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}} \tag{11}$$

A positive skill score indicates that the model outperforms persistence forecasting, while negative values indicate inferior performance.

To quantify predictive uncertainty in WQI forecasts, a bootstrap-based ensemble approach was employed. The procedure consisted of the following steps:

- Bootstrap resampling of the training dataset with replacement using 200 iterations
- Recursive multi-step forecasting to generate monthly WQI predictions for the year 2026

For each forecast horizon, the ensemble of bootstrap predictions was summarized using:

- Median forecast value
- 10th percentile, representing the lower uncertainty bound
- 90th percentile, representing the upper uncertainty bound

In addition to machine learning–based forecasting, time-series forecasting was performed using Tableau Public to provide a comparative baseline. Tableau uses an automated forecasting framework based on exponential smoothing (ETS) models, which decompose the time series into trend, seasonal, and residual components.

Forecasts were generated using Tableau's built-in forecasting functionality, which automatically selects the most appropriate exponential smoothing model based on data characteristics.

The forecast horizon was extended beyond the available observations to obtain projections up to the year 2026. Tableau also provides confidence intervals around the forecasted values, representing uncertainty estimates derived from the underlying statistical model.

Unlike the machine learning approach, Tableau forecasting relies solely on historical WQI values and does not incorporate lagged features, nonlinear interactions, or external drivers. This makes it suitable as a reference statistical baseline for comparison with data-driven machine learning models.

5 Results(Visualisations)

5.1 Spatial and Waterbody-wise Assessment of Water Quality in Kerala

5.1.1 Spatial Coverage of Data

Higher intensity corresponds to a greater number of recorded water bodies or monitoring locations.Among the districts, Kannur, Kasaragod, and Thiruvananthapuram recorded the highest number of water bodies (10 each).In terms of monitoring locations, Kannur had the highest count (29), followed by Ernakulam (22) and Kasaragod (17).

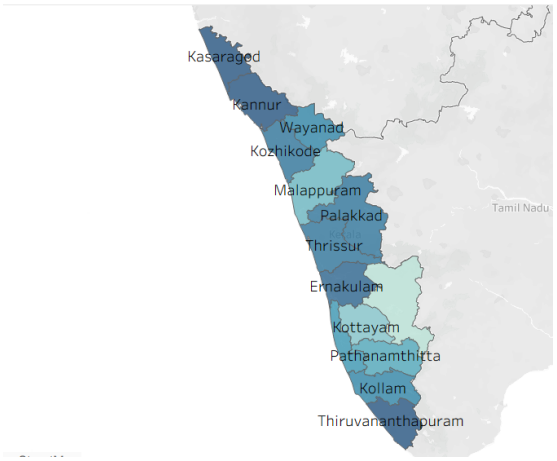


Figure 2: District-wise distribution of waterbodies in the dataset

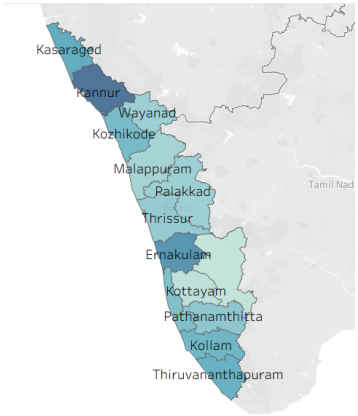


Figure 3: District-wise distribution of monitoring locations in the dataset

5.1.2 Assessment of River Water Quality Using Indian Drinking Water Standards and CCME Water Quality Index

The distribution of waterbodies across different water quality categories based on Indian WQI and CCME WQI is summarized below.

Table 4: Waterbodies distribution by water quality categories based on Indian WQI

WQI Category	Number of Waterbodies	Percentage (%)
Excellent quality	4	5.00
Good quality	53	66.25
Poor quality	12	15.00
Very poor quality	4	5.00
Unfit for drinking	7	8.75
Total	**80**	**100**

Table 5: Waterbodies distribution by water quality categories based on CCME WQI

WQI Category	Number of Waterbodies	Percentage (%)
Excellent quality	20	25.00
Good quality	4	5.00
Fair quality	9	11.25
Marginal quality	17	21.25
Poor quality	30	37.50
Total	**80**	**100**

Figures 4–6 show the waterbodies corresponding to the Excellent, Poor, and Unsuitable for Drinking WQI categories based on Indian WQI . While a majority of rivers (66.25%) exhibit good water quality, the presence of rivers categorized as poor, very poor, or unfit for drinking highlights the need for targeted monitoring and management interventions.But,in some rivers (e.g., Ashtamudi, Kayamkulam Kayal), reduced WQI values are primarily influenced by natural salinity, resulting in elevated conductivity and dissolved solids, rather than direct freshwater pollution.

Based on the CCME Water Quality Index, most of the monitored rivers (37.5%) fall under the poor water quality category, which is a matter of concern. Only 25% of the rivers exhibit excellent water quality, while the remaining sites are distributed across intermediate categories.

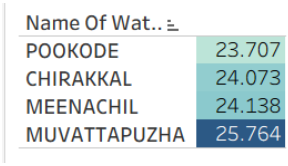


Figure 4: Waterbodies with Excellent Water Quality

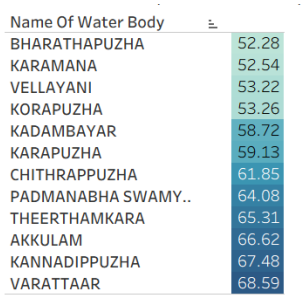


Figure 5: Waterbodies with Poor Water Quality

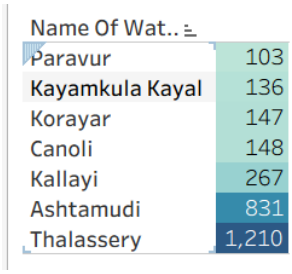


Figure 6: Unsuitable for Drinking

River-wise Comparison of Water Quality Across Monitoring Locations

Karamana River: Overall WQI Classification (Indian WQI-Poor; CCME WQI-Marginal)

The station-wise analysis of the Karamana River showed spatial variability in water quality. Among the four monitoring locations, two indicated good water quality under the Indian WQI , whereas Thiruvallom and Moonnattumukku were classified as poor and very poor, respectively. In the Indian WQI framework, Moonnattumukku showed higher WQI values, reflecting the influence of its coastal location. As a result, this station disproportionately affects the averaged Indian WQI, leading to the classification of the Karamana River as Poor (average WQI = 52.54), despite good quality at upstream locations.However,the CCME WQI

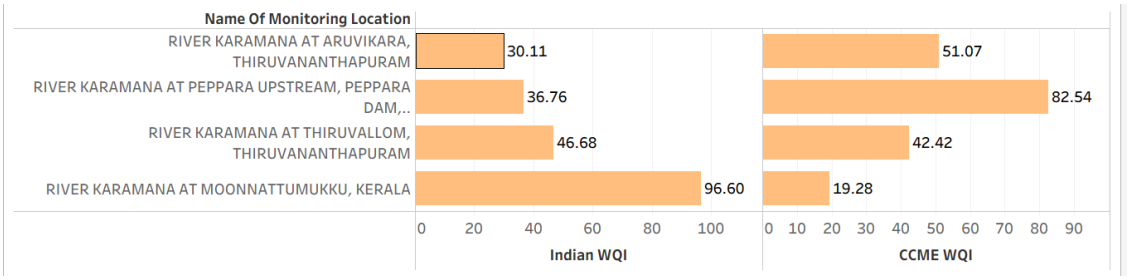


Figure 7: Comparison of Water Quality Index across monitoring locations for Karamana

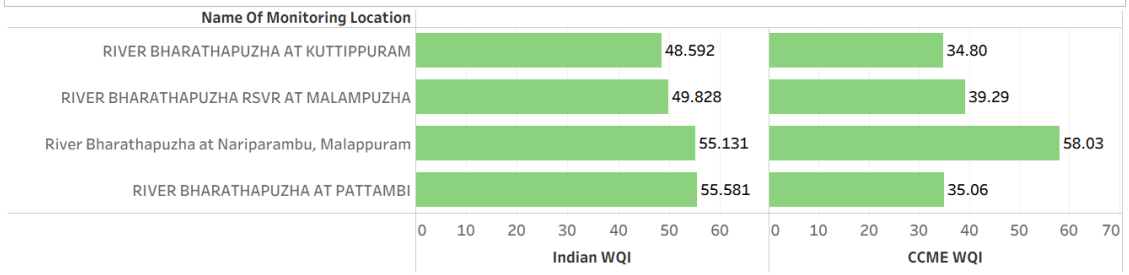


Figure 8: Comparison of Water Quality Index across monitoring locations for Bharathapuzha

gave a complementary perspective by assessing overall water quality rather than drinking-water suitability. Since salinity-related parameters such as total dissolved solids and conductivity are not included in the CCME WQI formulation used in this study, the low CCME WQI values observed at Moonnattumukku and Thiruvallom indicate degradation in general water quality that cannot be attributed to salinity effects. This suggests the possible influence of other non-salinity-related factors in these coastal stretches.

Bharathapuzha River:Overall WQI Classification (Indian WQI-Poor; CCME WQI-Poor)

In contrast, the Bharathapuzha River exhibits a relatively uniform spatial pattern in Indian WQI values, with monitoring locations generally ranging from 48 to 55. The river-level average Indian WQI (52.28) indicates an overall decline in drinking-water quality that is not restricted to specific stretches. Notably, inland stations such as Pattambi and Malampuzha, which are not influenced by coastal salinity, also record values close to the good–poor threshold values, suggesting a widespread concern along the river. Given its importance as a drinking water source, this highlights the need for basin-wide monitoring and management.The CCME WQI analysis similarly indicates general water quality degradation across the Bharathapuzha River. Three monitoring locations are classified as poor, while one location falls under the marginal category. This pattern suggests that water quality degradation is widespread rather than localized.The agreement between the Indian WQI and CCME WQI assessments suggests a basin-scale decline in water quality, which raises concern for drinking-water suitability and general ecological health.

Thalassery River:Overall WQI Classification (Indian WQI-Unfit for drinking; CCME WQI-Marginal

In the Thalassery River, Pathipalam and Kunduchira have good water quality, suitable for drinking, while other locations are unfit. Koduvalli shows an exceptionally high Indian WQI (4000), likely due to saline intrusion, which raises conductivity and dissolved solids. This localized effect skews the river's average Indian WQI, illustrating how spatially averaged indices can hide important intra-river variations.Under the CCME WQI assessment, the Thalassery River shows relatively better general water quality compared to the Indian WQI drinking-water classification. Three monitoring locations fall within the marginal category, while Koduvalli is classified as poor. Notably, although Koduvalli remains the weakest station, its CCME WQI value (39) indicates moderate degradation.Compared to the Indian WQI, which classified the river

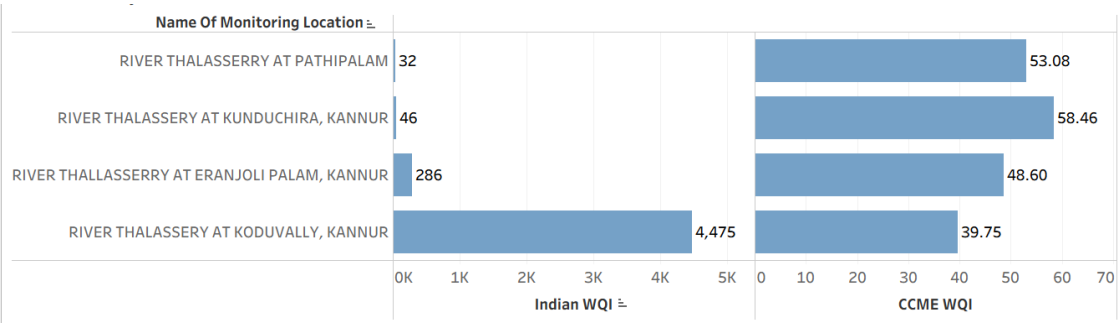


Figure 9: Comparison of Water Quality Index across monitoring locations for Thalassery River

as unfit for drinking,the CCME assessment classifies most locations as marginal, with Koduvalli as poor. This difference shows the sensitivity of drinking-water indices to salinity, while the CCME WQI provides a measure of general water quality that appears moderately reduced rather than severe degradation.

Pamba River:Overall WQI Classification (Indian WQI-Good; CCME WQI-Marginal)

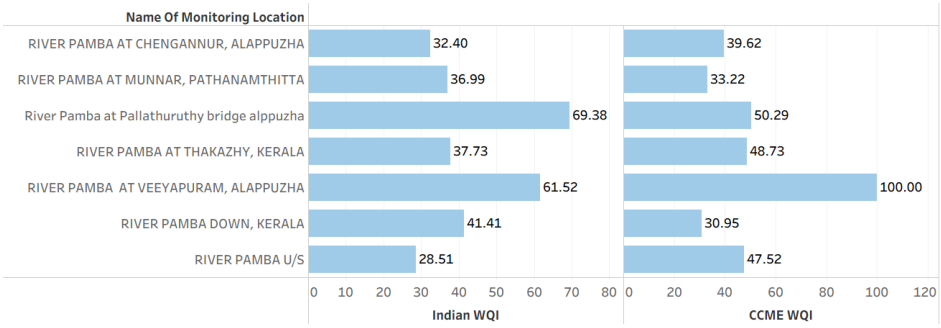


Figure 10: Comparison of water quality index across monitoring locations for pamba river

Although the Pamba River is classified as having good water quality based on its average Indian WQI, spatial variability is observed across monitoring locations. Most stations record WQI values in the range of 28–37, indicating good drinking water quality. However, downstream stations such as Pallathuruthy Bridge, Veeyapuram, and Pamba Down exhibit relatively higher WQI values, suggesting localized deterioration in drinking water quality.

Under CCME classification, three stations fall in the poor category, three in marginal, and only one in excellent. Notably, Veeyapuram, which shows poor drinking water quality under the Indian WQI, is classified as having excellent general water quality under CCME. Conversely, stations such as Chengannur, which exhibit good drinking water quality, show poor water quality under the CCME WQI.

This contrast highlights that locations with acceptable drinking water quality do not necessarily exhibit good overall ecological or general water quality.

5.2 Correlation Analysis

5.2.1 Pearson correlation analysis

The Pearson correlation heatmap (Figure 11) illustrates the linear relationships among water quality parameters. Strong positive correlations are observed among electrical conductivity, total dissolved solids, sodium, chloride, hardness, calcium, magnesium, and sulphate, indicating a salinity-controlled hydrochemical system.

Moderate correlations are observed among biochemical oxygen demand (BOD), chemical oxygen demand (COD), and nitrogen species, suggesting organic pollution and nutrient enrichment. Microbiological

parameters such as total coliform, fecal coliform, and fecal streptococci show strong mutual correlations.

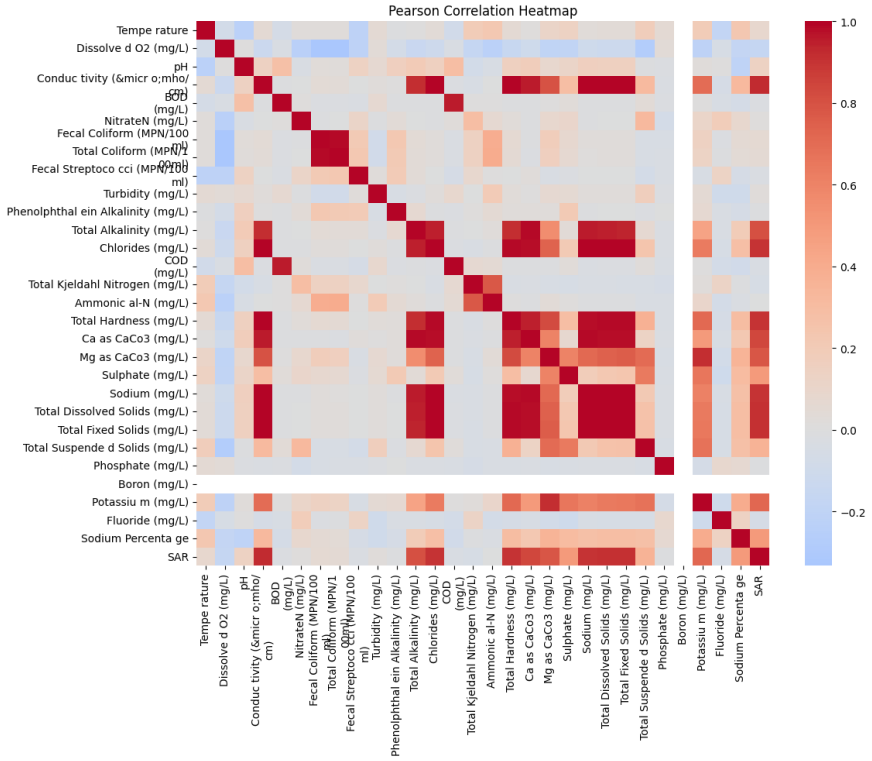


Figure 11: Pearson correlation heatmap showing linear relationships among physicochemical, nutrient, and microbiological parameters.

5.2.2 Principal Component Analysis

The PCA scree plot (Figure 12) shows that the first five principal components explain approximately 70% of the total variance, indicating that a limited number of dominant processes control water quality variability.

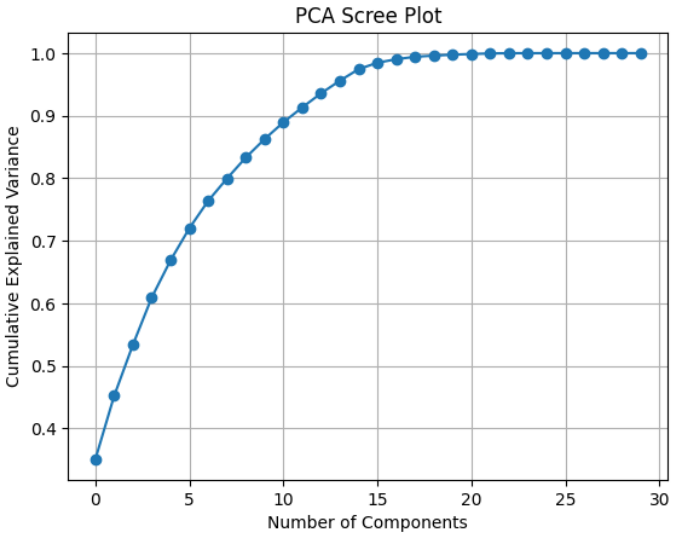


Figure 12: PCA scree plot showing cumulative explained variance of principal components.

5.2.3 Mutual information analysis

Figure 13 presents the mutual information scores of key water quality parameters. Microbiological indicators exhibit the highest scores, followed by hardness and major ion parameters, highlighting the importance of non-linear biological and geochemical influences on water quality.

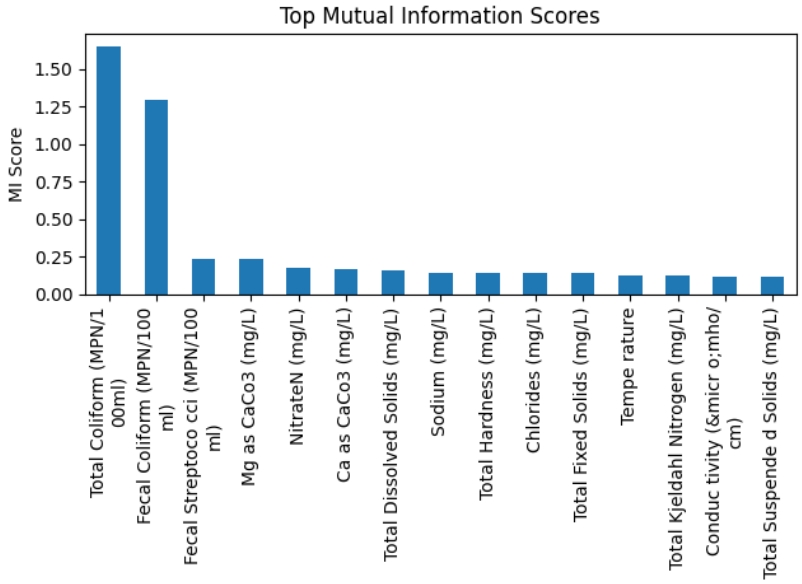


Figure 13: Top mutual information scores indicating non-linear dependencies among water quality parameters.

5.2.4 Random Forest feature importance

The Random Forest feature importance results (Figure 14) identify fecal coliform and total coliform as the most influential parameters governing overall water chemistry. Physicochemical parameters such as hardness, alkalinity, and conductivity show secondary importance.

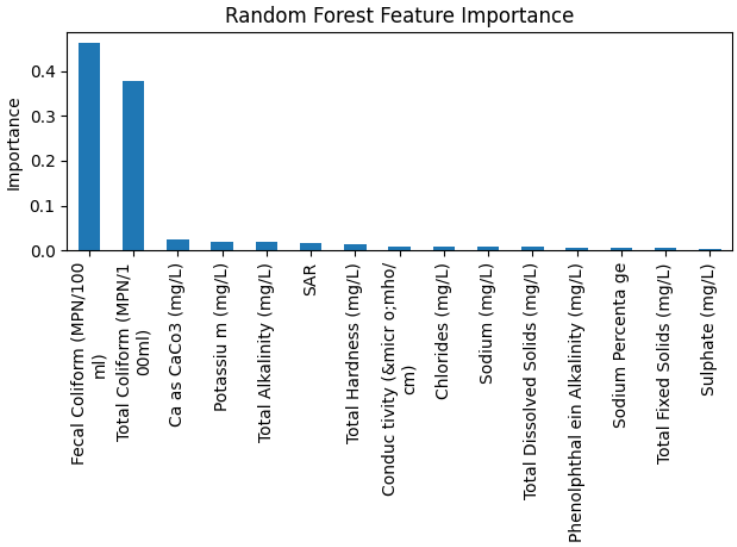


Figure 14: Random Forest feature importance ranking of water quality parameters.

5.3 Physical Analysis of Water

5.3.1 Colour

Kannur has more green colour rivers indicating high algae content, and Kasargod has more clear-water rivers.

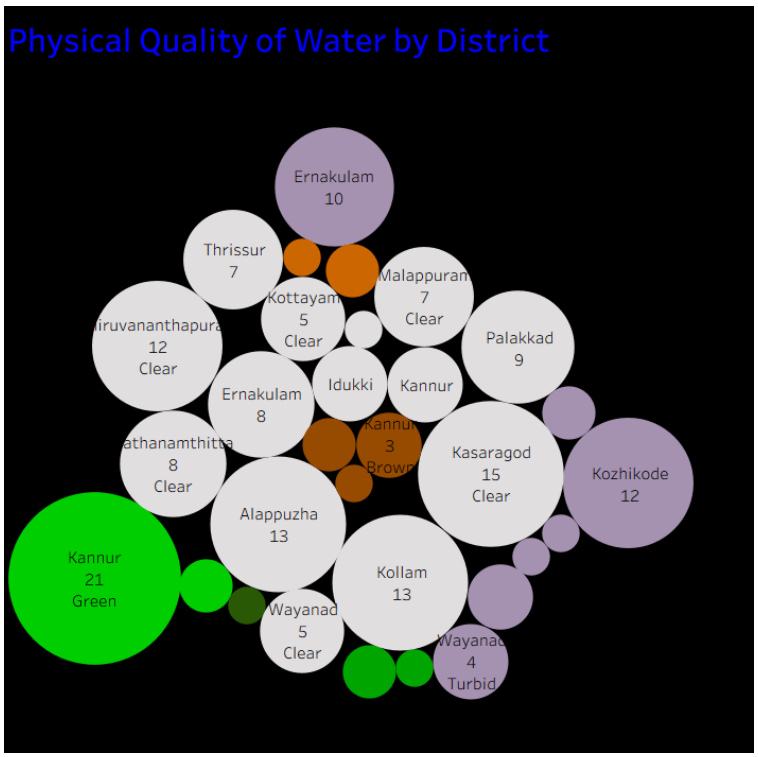


Figure 15: Colour Analysis

5.3.2 Temperature

Rivers in Thiruvananthapuram and Kottayam have the highest temperature, 29.35°C and 28.2°C respectively, which decrease the dissolved oxygen concentration, causing stress for aquatic life..

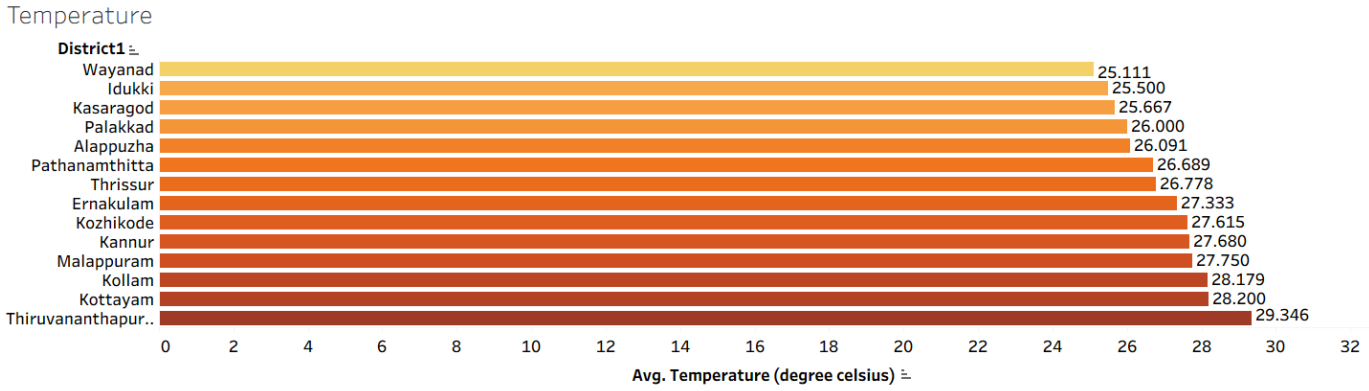


Figure 16: Average temperature in rivers

5.3.3 TDS

Kollam and Kannur have high TDS amounts. High TDS often indicates high levels of hardness or mineral content.

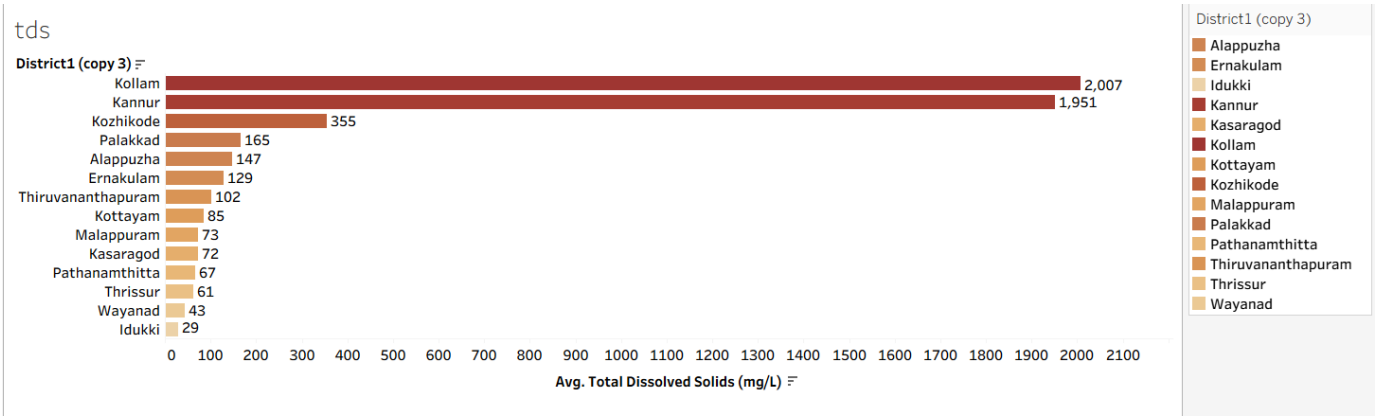


Figure 17: Total Dissolved Solids in water

5.3.4 Total Suspended Solids

Rivers in Kollam has the highest amount of suspended solids.

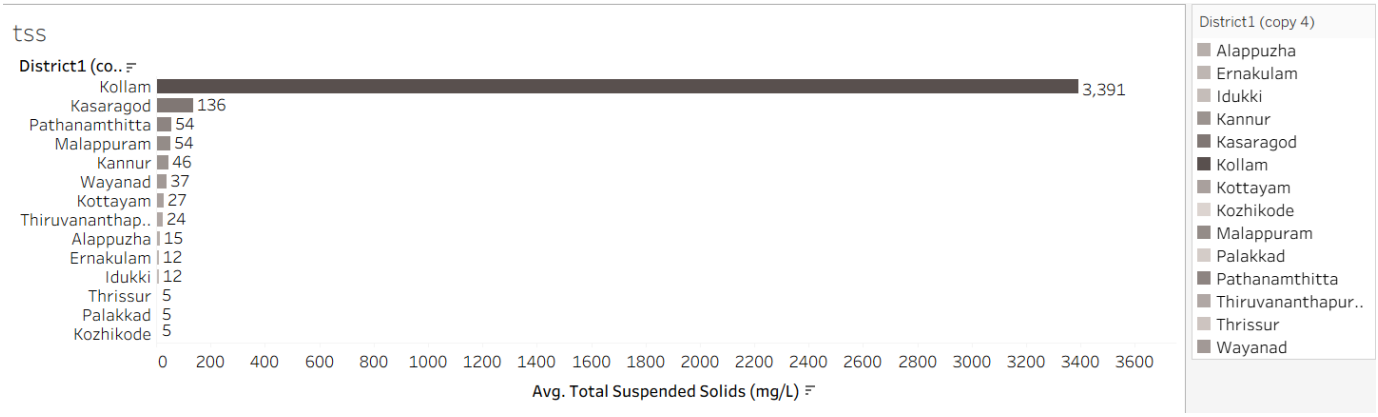


Figure 18: Total Suspended solids in water

5.4 Chemical Analysis of Water

5.4.1 pH

Typically, river water has a pH range of approximately 6.5 to 8.5. From the figure 19 , our analysis also lies in this range.

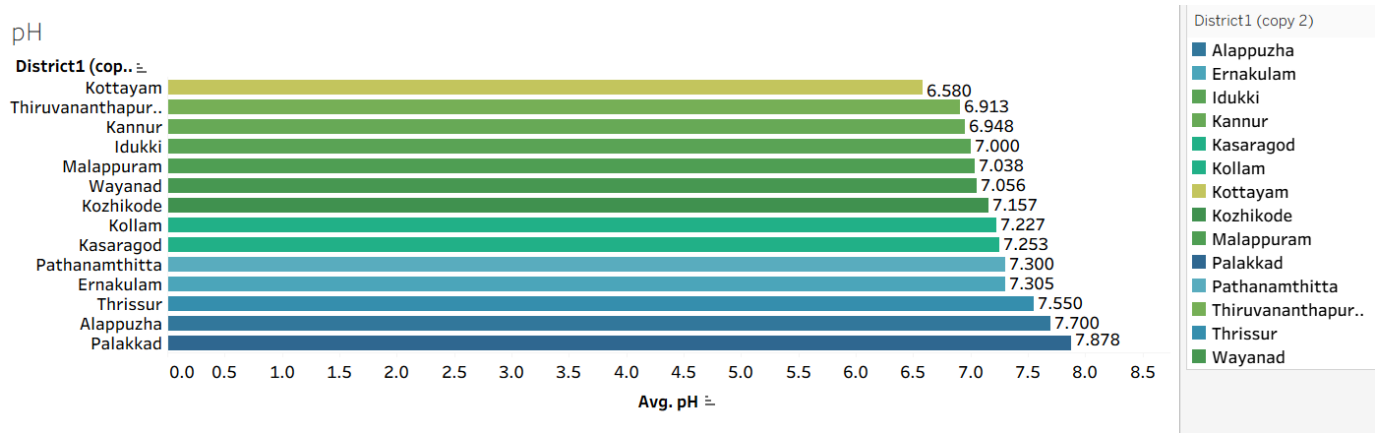


Figure 19: pH of river water

5.4.2 Dissolved Oxygen

Rivers in Kottayam and Thiruvanathapuram have less dissolved oxygen, which may be due to the high temperature in rivers (Figure 16).

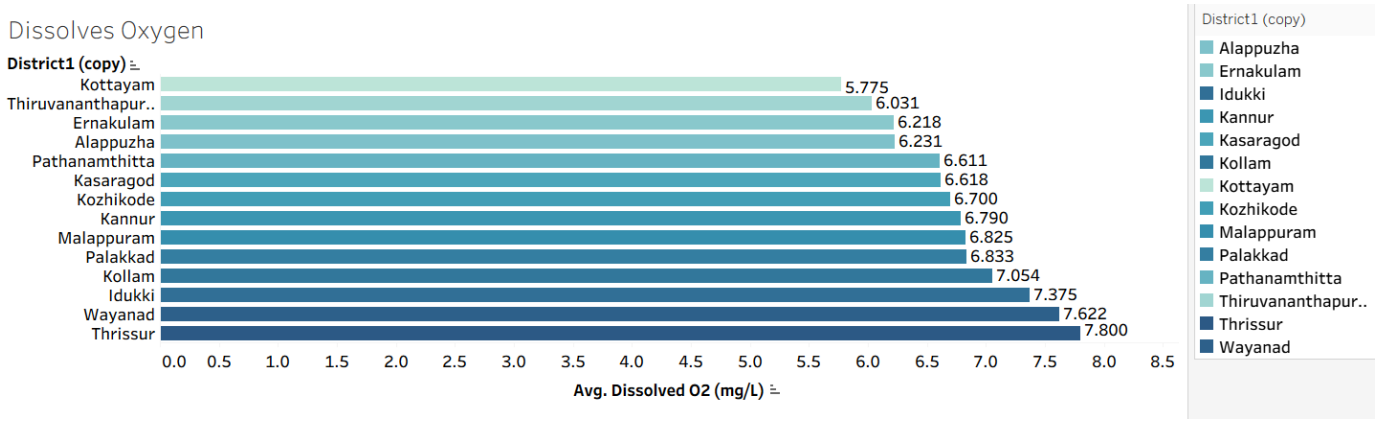


Figure 20: Average dissolved oxygen in rivers

5.4.3 Temporary Hardness

Kollam and Kannur have high temporary hardness, which may be due to high TDS content(Figure 17).

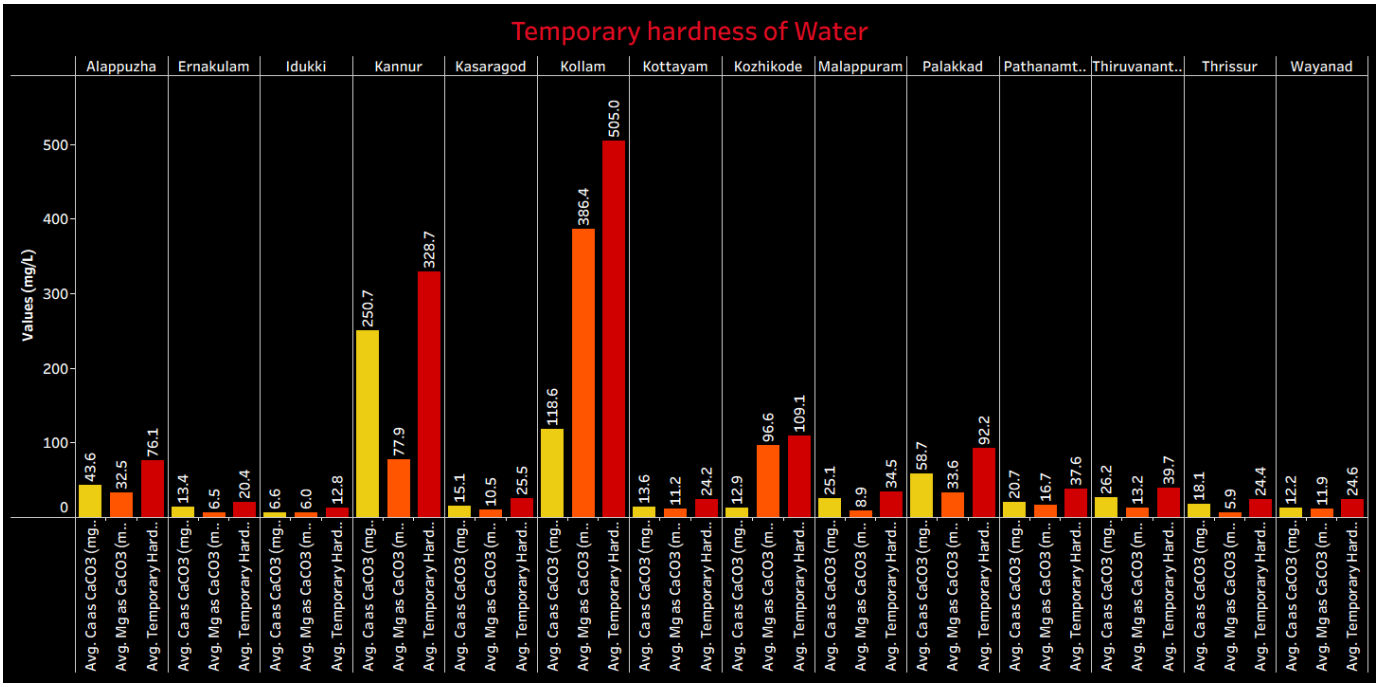


Figure 21: Temporary hardness of water

5.4.4 Nitrate Content

In clear water, the average nitrate content is 0.64, which is good. In dark green-coloured rivers, the nitrate content is very high, which causes the overgrowth of algae (Eutrophication), and thus the rivers appear green in colour.

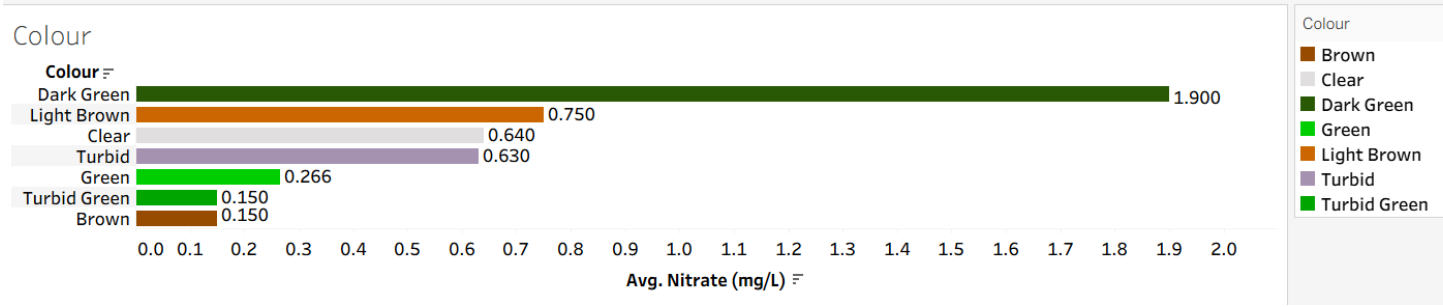


Figure 22: Nitrate content in rivers compared with the river colours

5.4.5 Biochemical Oxygen Demand

In general, BOD levels in water bodies are between 3-5 mg/L .From the figure we can observe that Kayamkulam kayal is having a BOD value of 6.30 mg/L,which is a bit higher than the general value ,which indicates significant pollution that can harm aquatic life.

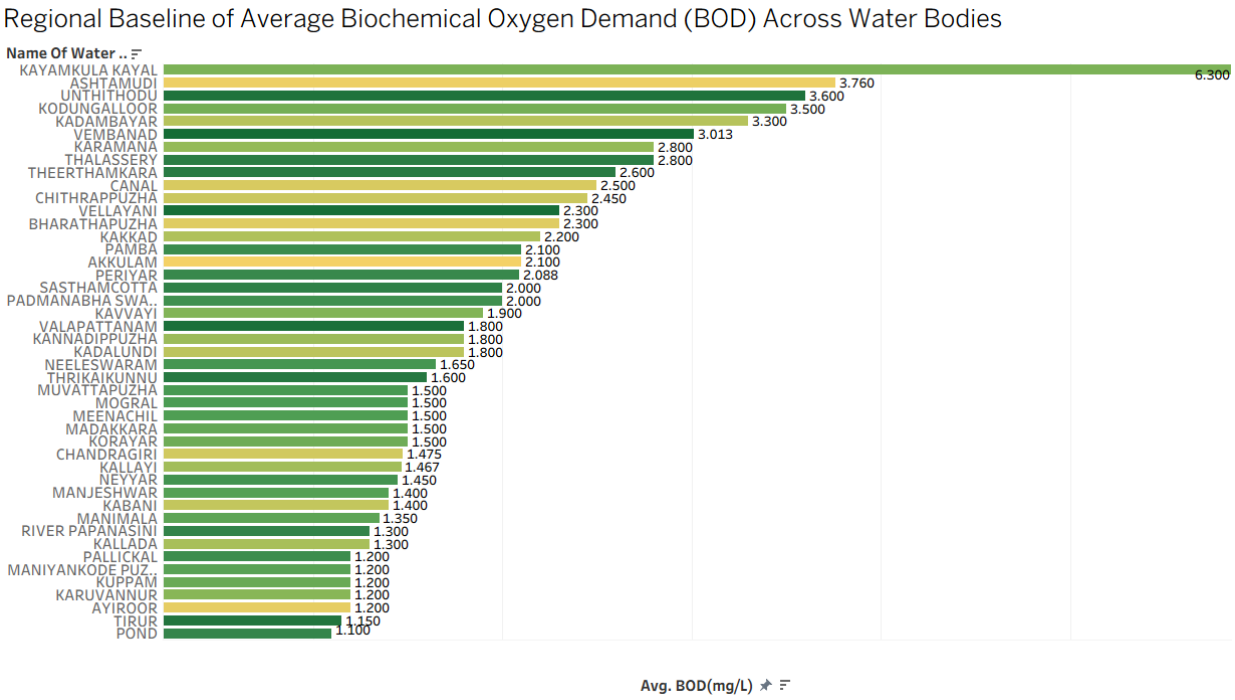


Figure 23: Regional baseline of average biochemical oxygen demand (BOD) across inland water bodies monitored under the National Water Monitoring Programme (NWMP).

5.5 Machine - Learning based WQI Forecast

5.5.1 Model Performance for 2025 Forecast

The predictive performance of the evaluated models was assessed with training data up to 2024 and testing data from 2025. Model accuracy was quantified using Root Mean Squared Error (RMSE), and performance was further evaluated relative to a persistence baseline using the skill score metric.

Forecast Performance for the Periyar River (2025)

Table 6 summarizes the forecasting performance of the evaluated models.

Table 6: Performance of regression models for 2025 WQI forecasting for the Periyar River.

Model	RMSE	Skill Score
XGBoost	126.66	0.591
Gradient Boosting	186.67	0.397
Support Vector Regression (SVR)	200.31	0.353
Linear Regression	210.80	0.319
Random Forest	213.41	0.311
Persistence Baseline	309.61	0.000

For the Periyar River, the historical WQI time series exhibits strong nonlinearity, abrupt fluctuations, and intermittent extreme values. In such a system, tree-based ensemble models—particularly XGBoost—outperformed other approaches, achieving a skill score of 0.59 relative to a persistence baseline. The superior performance of XGBoost can be attributed to its ability to capture nonlinear temporal dependencies and interactions between lagged WQI values and seasonal features. The wide and asymmetric uncertainty bands observed in the machine learning forecast reflect sensitivity to historical extremes and the propagation of uncertainty in recursive forecasting.

5.5.2 2026 WQI Forecast with Uncertainty for the Periyar River

Based on its superior validation performance, the XGBoost model was selected to generate monthly WQI forecasts for the Periyar River for the year 2026. Uncertainty was quantified using bootstrap resampling with 200 iterations and recursive multi-step forecasting.

Table 7 presents the median forecast along with the 10th and 90th percentile uncertainty bounds.

Table 7: Monthly WQI forecast for 2026 with uncertainty bounds for the Periyar River (XGBoost).

Month	Median	Lower 10%	Upper 90%
Jan-26	185.41	36.76	642.53
Feb-26	531.86	191.53	1272.03
Mar-26	650.16	395.30	1050.90
Apr-26	780.21	333.26	1114.45
May-26	343.84	41.41	806.04
Jun-26	111.35	33.77	177.73
Jul-26	66.85	34.00	125.55
Aug-26	69.35	36.47	202.32
Sep-26	52.41	32.63	123.54
Oct-26	60.63	31.78	181.79
Nov-26	62.26	41.22	94.51
Dec-26	127.37	47.65	151.55

The Periyar River forecast exhibits strong seasonal variability, with higher predicted WQI values during the pre-monsoon period and lower values during monsoon and post-monsoon months. The relatively wide uncertainty bands during early 2026 indicate sensitivity to historical variability and lagged system dynamics.

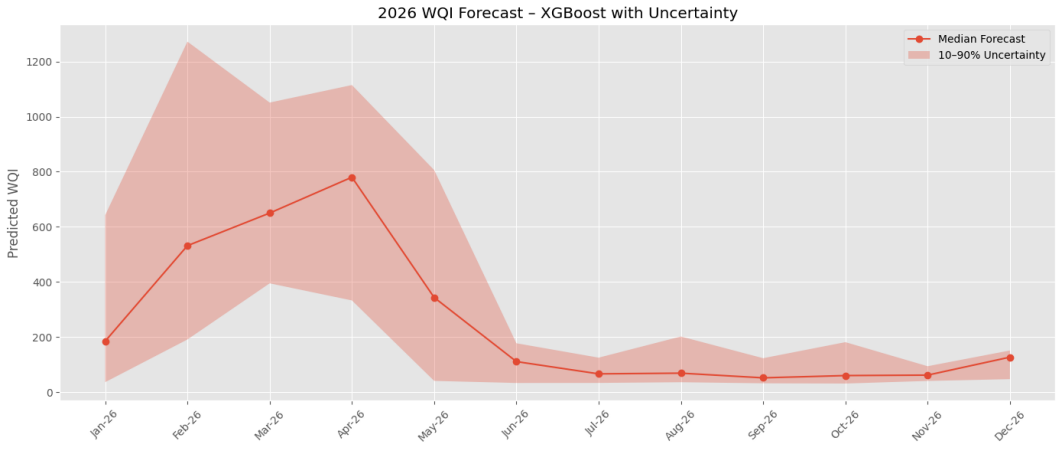


Figure 24: Machine learning–based WQI forecast for the Periyar River using the XGBoost model. The solid line represents the median forecast for 2026, while the shaded region denotes the 10–90% uncertainty interval obtained through bootstrap resampling.

5.5.3 Forecast Performance for the Bharathapuzha River (2025)

A similar strategy was applied to the Bharathapuzha River dataset. Table 8 summarizes the performance of the evaluated models for 2025.

For the Bharathapuzha River, the historical WQI series is comparatively smoother and more stable. In this case, Support Vector Regression (SVR) achieved the best predictive performance, with a skill score of 0.55. The strong performance of SVR, along with relatively good results from linear regression, suggests that Bharathapuzha exhibits more gradual temporal evolution and weaker nonlinear effects.

Table 8: Performance of regression models for 2025 WQI forecasting for the Bharathapuzha River.

Model	RMSE	Skill Score
Support Vector Regression (SVR)	9.97	0.552
Random Forest	10.61	0.524
Linear Regression	10.75	0.517
XGBoost	11.22	0.496
Gradient Boosting	11.68	0.476
Persistence Baseline	22.28	0.000

5.5.4 2026 WQI Forecast with Uncertainty for the Bharathapuzha River

The SVR model was therefore selected to generate the 2026 WQI forecast for the Bharathapuzha River. Bootstrap-based uncertainty quantification was applied following the same procedure as for the Periyar River. Table 9 presents the median forecast values along with the 10th and 90th percentile uncertainty bounds.

Table 9: Monthly WQI forecast for 2026 with uncertainty bounds for the Bharathapuzha River (SVR).

Month	Median	Lower 10%	Upper 90%
Jan-26	67.05	59.41	76.82
Feb-26	67.35	59.28	75.11
Mar-26	68.27	59.76	78.12
Apr-26	68.05	59.78	75.63
May-26	68.35	59.64	76.65
Jun-26	68.15	59.70	76.24
Jul-26	68.68	59.49	76.58
Aug-26	68.48	59.23	75.88
Sep-26	68.79	59.34	79.25
Oct-26	68.43	59.28	76.48
Nov-26	68.91	59.47	80.26
Dec-26	68.74	59.45	77.08

The Bharathapuzha River forecasts exhibit comparatively narrower uncertainty bands, indicating greater predictability and less extreme temporal variability relative to the Periyar River. This suggests differences in hydrological regulation, pollution loading, and response times between the two river systems.

5.5.5 Comparative Interpretation

The contrasting model performance across the two rivers highlights the importance of river-specific properties in water quality forecasting. While tree-based ensemble methods such as XGBoost performed best for the Periyar River, kernel-based SVR models were more effective for the Bharathapuzha River. These differences shows that no single model is universally optimal and that customised strategies are essential for reliable river-specific water quality prediction.

5.5.6 Tableau-Based Forecast Results for the Periyar River

Figure 26a shows the Tableau-generated WQI forecast for the Periyar River. The historical WQI series exhibits pronounced variability, with sharp peaks and rapid declines, indicating strong fluctuations in water quality with time.

The Tableau forecast extends this trend into the future with a smooth continuation of recent patterns. The associated confidence interval widens significantly toward the forecast horizon, reflecting increased uncertainty due to high historical variability and limited data in recent periods.

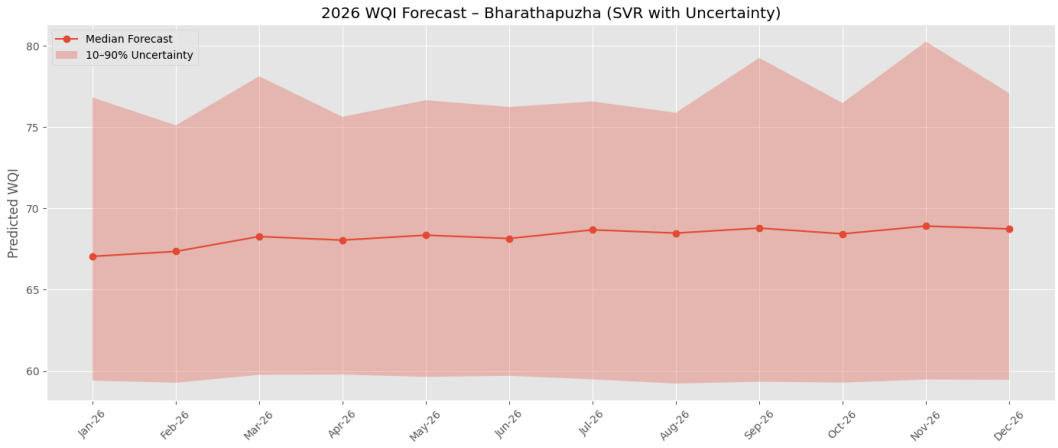
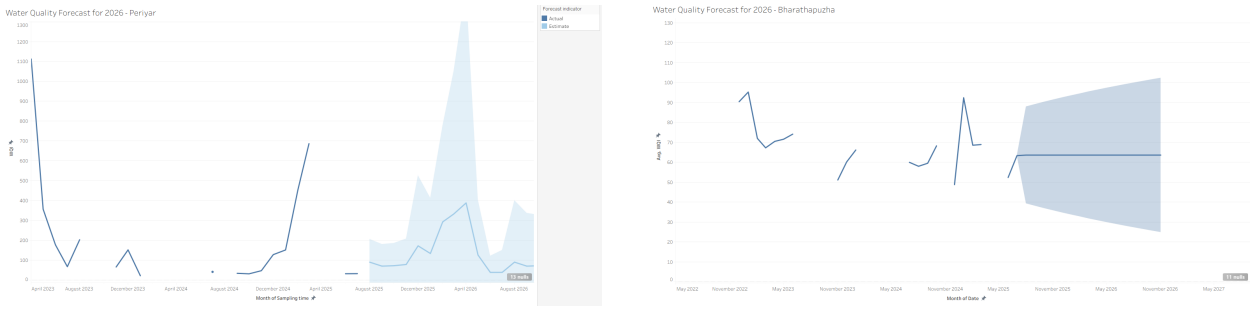


Figure 25: Machine learning–based WQI forecast for the Bharathapuzha River using Support Vector Regression (SVR). The solid line represents the median forecast for 2026, while the shaded region denotes the 10–90% uncertainty interval obtained through bootstrap resampling.

The forecast suggests moderate WQI values in the near future, but the broad uncertainty bounds indicate limited confidence in precise point predictions. This behavior is characteristic of exponential smoothing models when applied to highly non-stationary environmental time series.



(a) Periyar River

(b) Bharathapuzha River

Figure 26: Tableau-generated WQI forecasts showing historical values, forecasted trends, and confidence intervals for (a) Periyar River and (b) Bharathapuzha River.

5.5.7 Tableau-Based Forecast Results for the Bharathapuzha River

The Tableau forecast for the Bharathapuzha River is shown in Figure 26b. Compared to the Periyar River, the historical WQI series for Bharathapuzha exhibits smoother temporal variations.

As a result, the Tableau forecast displays a relatively stable projected WQI trend with narrower confidence intervals. The forecast indicates near-constant WQI values over the year 2026.

5.5.8 Comparison Between Tableau and Machine Learning Forecasts

A comparison between Tableau-based forecasts and machine learning–based forecasts reveals important differences in predictive behavior, uncertainty representation, and interpretability.

For the Periyar River, machine learning models—particularly XGBoost—demonstrated superior performance in 2025 validation, with a skill score of 0.59 relative to the persistence baseline. The ML-based 2026 forecasts exhibited strong seasonal structure and asymmetric uncertainty bounds, indicating nonlinear temporal dynamics and error propagation captured through lagged features and bootstrap resampling. In contrast, the Tableau forecast may underestimate the complexity of the underlying system.

For the Bharathapuzha River, both Tableau and machine learning approaches produced relatively stable forecasts. However, the SVR-based machine learning forecast provided tighter uncertainty bounds and slightly improved predictive skill compared to Tableau's exponential smoothing model. This indicates that even in systems with smoother dynamics, machine learning models can better exploit temporal dependencies.

Overall, Tableau forecasting serves as a useful statistical baseline but lacks the flexibility to capture nonlinear relationships in river water quality dynamics. Machine learning approaches, augmented with lag-based features and uncertainty quantification, offer improved accuracy, interpretability, and robustness for operational forecasting applications.

6 Conclusion

This study evaluates river water quality across Kerala using a combination of statistical analyses, multivariate techniques, machine learning methods, and water quality indices. The assessment focuses on identifying the dominant factors controlling water chemistry, quantifying biological and geochemical influences, and examining spatial and river-wise variability in water quality. Drinking-water suitability and general water quality were assessed using the Indian Weighted Arithmetic Water Quality Index and the CCME Water Quality Index, while forecasting approaches were applied to understand temporal behavior in selected rivers. The results give insights into the water quality conditions of Kerala and their variability across districts and river systems.

The strong correlations observed among conductivity, dissolved solids, major ions, and hardness parameters indicate that salinity-driven processes such as mineral dissolution and ionic enrichment play a key role in controlling water chemistry. These relationships are characteristic of water systems influenced by geogenic inputs and anthropogenic activities.The dominance of microbiological parameters in both mutual information and Random Forest analyses suggests that biological contamination is a major factor influencing overall water quality variability. This may be attributed to sewage discharge, agricultural runoff, or inadequate sanitation practices within the study area.The agreement between statistical correlation, PCA, mutual information, and machine learning analyses strengthens the reliability of the findings. Overall, the results indicate that water quality is primarily governed by biological contamination, with secondary control from salinity-related geochemical processes and nutrient enrichment.

Although most waterbodies are classified as having good quality under the Indian WQI, the occurrence of rivers rated as poor, very poor, or unfit for drinking signifies the need for focused monitoring and management efforts. In some waterbodies, reduced drinking-water quality is primarily driven by natural salinity intrusion, emphasizing the sensitivity of drinking-water indices to conductivity and dissolved solids. In contrast, the CCME WQI classified a significant proportion of waterbodies as poor, indicating broader concerns regarding general water quality and ecological health. River-wise analysis further revealed that spatial averaging can mask important intra-river variability. In rivers such as Karamana and Thalassery, localized degradation at specific monitoring locations disproportionately influenced overall classifications, while in Bharathapuzha, consistently reduced values across stations indicates basin-scale deterioration. Conversely, the Pamba River demonstrated that acceptable drinking-water quality does not necessarily imply good general water quality,which shows the importance of multi-index assessments.Because sampling took place during the monsoon, the results capture short-term fluctuations in water quality rather than stable long-term conditions.Overall, the findings demonstrate that integrating drinking-water-focused and general water quality indices, together with location specific monitoring, is essential for accurately identifying degradation hotspots and informing effective river management and policy decisions in Kerala.

The overall analysis of physicochemical properties of waterbodies in Kerala across the district shows spatial variations influenced by temperature, dissolved solids and nutrient levels. Elevated water temperatures observed in rivers of Thiruvananthapuram and Kottayam are associated with reduced dissolved oxygen levels, which can impose stress on aquatic organisms. High TDS in rivers of Kollam and Kannur shows increased mineral content and hardness, which is supported by the high temporary hardness recorded in these areas. Rivers in Kollam also show the highest suspended solid content pointing to greater turbidity and possible sediment content. The pH values across all rivers fall within the normal range of 6.5 to 8.5 indicating no extreme acidity or alkalinity issues. And also the rivers in Kannur exhibits a green colour due to excess algae growth which is caused by high nitrate content (eutrophication).

ML based forecasting used machine learning techniques to analyze and predict river water quality in Kerala, with a focus on the Periyar and Bharathapuzha rivers. The results demonstrate that while both Tableau-based exponential smoothing and machine learning approaches provide useful insights, their predictive behavior, uncertainty representation, and interpretability differ substantially depending on the underlying river dynamics.

The comparison between the two rivers highlights that no single forecasting approach is universally optimal. River-specific hydrological, geochemical, and anthropogenic influences strongly affect model suitability. Machine learning models, particularly when combined with lag-based features and bootstrap uncertainty quantification, provide greater flexibility and realism in representing complex systems. Tableau-based forecasting, on the other hand, serves as a useful statistical baseline and exploratory tool, especially for systems with smoother temporal behavior.

Future studies can improve this analysis by including information on land use, population density, and known pollution sources, which would help identify the specific causes of water quality degradation. Overall, combining visual analysis, statistical methods, and machine learning provides a clearer and more reliable understanding of water quality patterns. The findings also show that Physicochemical properties of water is worsening in several districts, highlighting the need for focused management actions and effective pollution control to safeguard Kerala's water resources.

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